

Smart Irrigation Using AI in Agriculture

Watering Crops... Not Wasting Drops!

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Problem Statement

- Over-irrigation:
 - Water wastage
 - Soil nutrient loss
- Under-irrigation:
 - Poor crop growth
- Challenges:
 - Unpredictable weather
 - Lack of real-time monitoring
 - Manual irrigation decisions
- Research Gap:
 - Need for intelligent irrigation systems

Funny Analogy: Traditional irrigation is like leaving your shower on all day!

- Data Collection:
 - Soil moisture sensors
 - Weather data
- Data Preprocessing:
 - Cleaning and normalization
- Apply AI/ML models
- Generate irrigation schedule
- Automate water delivery

Research Perspective:

- Time-series prediction
- Decision-making systems

Funny Analogy: Like giving plants a smartwatch to track their thirst!

Data Description

- Input Features:
 - Soil moisture
 - Temperature
 - Humidity
 - Rainfall
 - Soil type
- Data Sources:
 - IoT sensors
 - Weather APIs
 - Historical farm data
- Output:
 - Irrigation decision (when and how much water)

Funny Analogy: Data is like a doctor's report—you shouldn't guess without it!

Model / Approach

- Machine Learning:
 - Decision Trees
 - Random Forest
- Deep Learning:
 - Artificial Neural Networks (ANN)
 - LSTM (time-series)
- Hybrid systems (rules + AI)

Why These Models?

- Adapt to changing environments
- Predict future water needs

Funny Analogy: LSTM remembers weather like your grandma remembers every rainy day!

- Water savings: 20% – 50%
- Improved crop yield
- Reduced labor effort
- Efficient resource usage

Evaluation Metrics:

- Water use efficiency
- Prediction accuracy
- Crop productivity

Funny Analogy: AI treats every drop like it's expensive coffee!

Conclusion

- Enables precision irrigation
- Reduces water wastage
- Enhances sustainability
- Supports smart agriculture

Future Scope:

- Mobile app integration
- Fully automated irrigation
- Climate-adaptive AI models

Funny Analogy: Future farms will water crops smarter than we water houseplants!

Questions?

Ask anything—even “Can AI water my plants when I forget?”

Thank You!